

Nazmul Islam Ananto

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EDUCATION

Bangladesh University of Engineering and Technology (BUET)

Dhaka, Bangladesh

BSc in Computer Science and Engineering

Apr. 2019 – Jul. 2024

- **CGPA: 3.91** ; Dean's List Award
- Notable Courses: Machine Learning, Compiler, Computer Networking, Security, AI, Architecture, Operating Systems, Data Structure & Algorithm

RESEARCH

COMPASS: Geo-Spatial Reasoning for Popular Path Query using LLMs

Undergraduate Thesis — supervised by Mohammed Eunus Ali & Md Rizwan Parvez

- Proposed **COMPASS**, a novel framework that uses Chain-of-Thought prompting with LLMs to tackle geo-spatial reasoning problems, particularly identifying and generating popular paths from historical trajectory data
- Designed a *Search* stage for extracting popular paths and a *Generate* stage for synthesizing novel routes under user-defined constraints — both leveraging LLMs' spatial reasoning and graph comprehension
- Demonstrated superior performance on real and synthetic datasets, derived from prior work, especially in sparse or constraint-based scenarios where traditional models underperform
- Evaluated using GPT-4o, GPT-o3-mini and Llama 3.1 8b, used OpenAI APIs; implementation to be open-sourced
- Targeting submission to AAAI 2026

EXPERIENCE

Software Engineer

Mar. 2024 – Present

Zelf Technologies Inc.

Brooklyn, NY, USA

- Built data pipelines collecting and processing 80M+ daily video analytics from social media public APIs
- Developed automation tools serving 100+ enterprise clients
- Collaborated with retail partners and cross-functional teams for critical features
- **Tech Stack:** Python, Selenium, Playwright, Appium, Burp Suite, Mitmproxy, ADB, GCP, Docker, GitHub Actions, Terraform

Research Assistant

Jul. 2022 – Jun. 2024

Institute of Water and Flood Management, BUET

Dhaka, Bangladesh

- Practical implementation of an Early Warning System for River Erosion prone areas
- Published in *Frontiers: A Riverbank Erosion Early Warning System*
- Led the Full-stack development team using React, Express, Firebase & PostgreSQL

Machine Learning Intern

May 2023 – Jun. 2023

Era-InfoTech Ltd

Dhaka, Bangladesh

SCORES & ACHIEVEMENTS

TOEFL iBT: 113/120 (R-30 L-29 S-27 W-27)

1st in Zelf Hackathon 1.0 at Dhaka – 2024

5th in IEEE GameJam at BUET – 2023: created *Asteroids* game with p5.js [Play Now]

PROJECTS

Stock Trader | *ML, Python*

- Automated trading agent trained on Dhaka Stock Exchange data for maximum profit

Maps 'n Bags | *ExpressJS, PostgreSQL, Firebase, Selenium*

- Itinerary planner with trip suggestions, customization, travel tracking & budgeting with synced data & distances from Google Maps

Creative Production Management | *System Design, BPMN, UI Mockups*

- Estimation & production system for a creative agency

Withered Away | *ExpressJS, OracleDB*

- Centralized management system for a retirement home's member & facilities

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL, C++, Bash, x86 Assembly

Frameworks: Flask, FastAPI, React, Express, NestJS, Apollo, Prisma

Tools: Selenium, Playwright, Appium, Git, Docker, Kubernetes, Terraform, GCP, Firebase, Linux Shell

Libraries & ML: NumPy, Pandas, LangChain, OpenCV, PyTorch, scikit-learn, NLTK, Matplotlib, Seaborn

Databases: PostgreSQL, MongoDB, MySQL, Oracle, Firebase Realtime DB

EXTRACURRICULAR

Director of Marketing & Communication

Jun. 2023 – Apr. 2024

BUET Cyber Security Club

Dhaka, Bangladesh

- Promoted cybersecurity events & hosted inter-university CTF (Catch The Flag) contests

President of Writing & Publication

Feb. 2018 – Feb. 2019

Notre Dame English Club

Dhaka, Bangladesh